

Prevalence and outcomes of patients experiencing in-hospital cardiac arrest in intensive care units: A systematic review and meta-analysis

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REVIEW TITLE AND BASIC DETAILS

Review title

Prevalence and outcomes of patients experiencing in-hospital cardiac arrest in intensive care units: A systematic review and meta-analysis

Condition or domain being studied

Cardiac Arrest; Intensive Care

Rationale for the review

To date, there exist no SRMAs performed evaluating the prevalence of IHCA while patients are in ICU. Correspondingly, there exist little literature on outcomes of these patients

Review objectives

What is the prevalence of patients who experience an in-hospital cardiac arrest while in the ICU ?

What are their outcomes - in terms of survival and functional neurological outcomes ?

Keywords

In-hospital cardiac arrest; ICU; Critical care

Country

Singapore; Australia

ELIGIBILITY CRITERIA

Population

Included

All retrospective or prospective observations and RCTs that reported on IHCA in adult patients (age > 18 years) admitted to the ICU will be included.

Excluded

We will exclude case reports, studies with palliative patients or patients admitted for organ donation.

We will exclude paediatric populations (18 years or younger) and animal studies.

Intervention(s) or exposure(s)

Included

Cardiac Care; Intensive Care

Comparator(s) or control(s)

Included

PICO tags selected: Intensive Care

The comparator group will comprise of patients in the ICU who do not experience an IHCA

Study design

Both randomized and nonrandomized study types will be included.

Included

RCTs, Cohort studies

Excluded

Case control studies, case report studies, cross-sectional studies, abstracts, reviews, meta-analyses

Context

Studies performed in the ICU setting and reporting on prevalence/outcomes of patients who experience an in-hospital cardiac arrest while in ICU

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO

01 June 2025

Review timeline

Start date: 1 June 2025. End date: 31 October 2025.

Date of registration in PROSPERO

26 July 2025

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol

A full protocol has not been written.

SEARCHING AND SCREENING

Search for unpublished studies

Only published studies will be sought.

Main bibliographic databases that will be searched

The main databases to be searched are *Embase.com*, *MEDLINE* and *Scopus*.

Search language restrictions

There are no language restrictions.

Search date restrictions

There are no search date restrictions.

Other methods of identifying studies

Other studies will be identified by: *contacting authors or experts* and *reference list checking*.

Link to search strategy

A full search strategy is not available.

Selection process

Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

Other relevant information about searching and screening

None

DATA COLLECTION PROCESS

Data extraction from published articles and reports

Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will not be contacted for further information.

Study risk of bias or quality assessment

Risk of bias will be assessed using: *Cochrane RoB-2* and *Newcastle-Ottawa*

Data will be assessed by one person (or a machine) and checked by at least one other person (or machine).

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment

Risk of bias due to missing results will be assessed

Certainty assessment

GRADE assessment will be used to review the certainty of evidence

OUTCOMES TO BE ANALYSED

Main outcomes

Prevalence of patients experiencing IHCA while in the ICU

Additional outcomes

Mortality

Functional recovery

Length of stay in ICU

Support use (vasoactive agents, mechanical ventilation, ECMO, KRT etc)

PLANNED DATA SYNTHESIS

Strategy for data synthesis

We will use R studio for the analysis of results.

Our primary outcome of prevalence will be measured using a generalised linear mixed model method. If there is significant inter-study heterogeneity, we will use a random-effects model to account for this inter-study heterogeneity. Secondary outcomes will also be pooled using GLMM. Where possible, we will make comparisons with the comparator group and present our results using odds ratios. We will use a p-value of <0.05 as significant. All analysis will be presented with their respective effect measures and corresponding 95% CIs.

CURRENT REVIEW STAGE

Stage of the review at this submission

Review stage

Started

Completed

Pilot work ✓
Formal searching/study identification
Screening search results against inclusion criteria
Data extraction or receipt of IPD
Risk of bias/quality assessment
Data synthesis

Review status

The review is currently planned or ongoing.

Publication of review results

Results of the review will be published in English.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

Mr Oliver Lim. National University of Singapore. Singapore.

No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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Review affiliation

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Funding source

Review has no funding and no agreed support from an academic institution and is done in authors' own time.

Peer review

There has been no peer review of this planned review.

ADDITIONAL INFORMATION

Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Medical Subject Headings

Cardiopulmonary Resuscitation; Heart Arrest; Hospitals; Humans; Intensive Care Units; Prevalence

SIMILAR REVIEWS

Check for similar records already in PROSPERO

PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 1 June 2025). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.

- Evaluating the incidence and clinical outcomes of Intensive care unit patients experiencing in-hospital cardiac arrest. A systematic review and meta-analysis. [published 1 August 2022] [CRD42022348124]. The review was acknowledged as **similar** but the authors opted to continue because *there are differences in intervention or comparator, there are differences in population, the review will be more up to date.* The authors also noted: *"The above study compares outcomes with patients in the general ward. Our review chooses to focus solely on ICU patients, and compares patients who experience an IHCA vs those who do not."*
- In-hospital cardiac arrest in ICU versus non-ICU patients with COVID-19 [published 15 December 2020] [CRD42020226262]. The review was judged **not to be similar**
- Pre- and intra-arrest prognostic factors for survival in in-hospital cardiac arrest – a systematic review and meta-analysis [published 9 August 2018] [CRD42018104795]. The review was judged **not to be similar**

PROSPERO version history

- [Version 1.0, published 26 Jul 2025](#)

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